

The kidney has

Two surfaces:

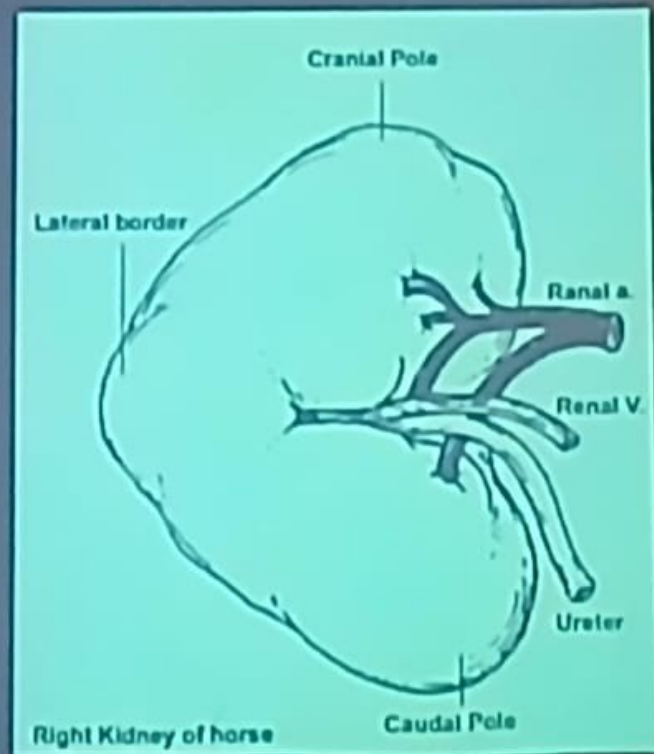
- convex dorsal & ventral surfaces.

Two borders:

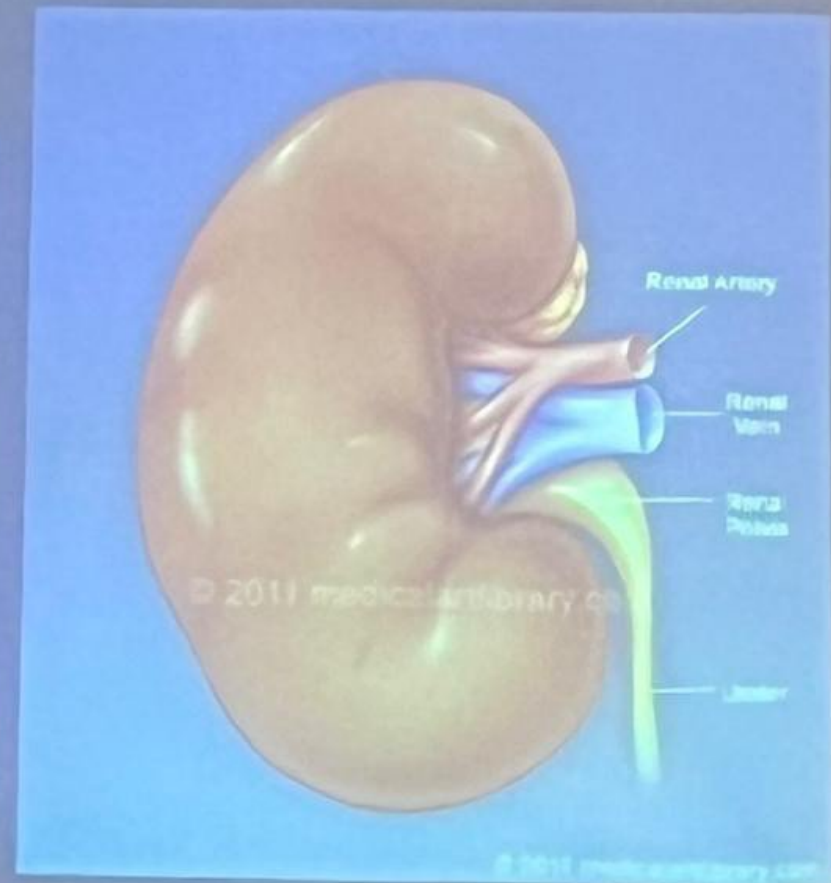
- straight medial border contain renal hilus.
- convex lateral border

Two poles:

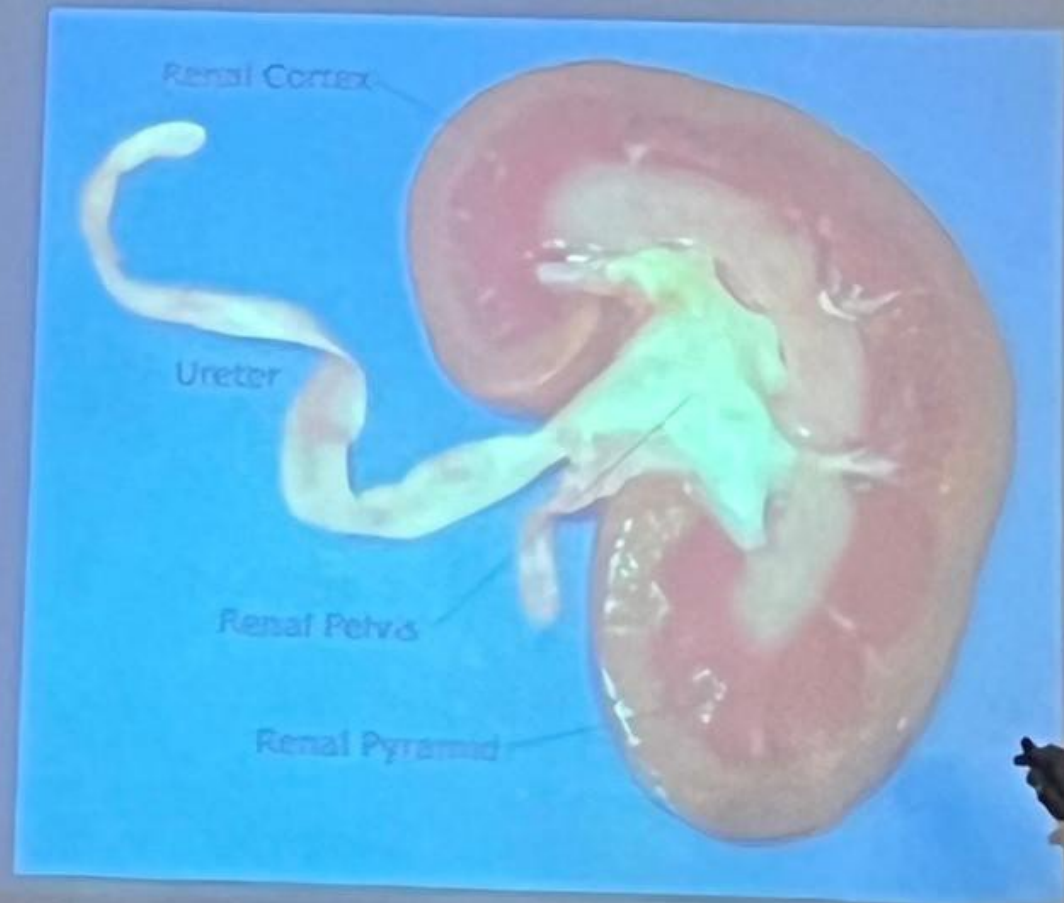
- cranial pole & caudal pole.



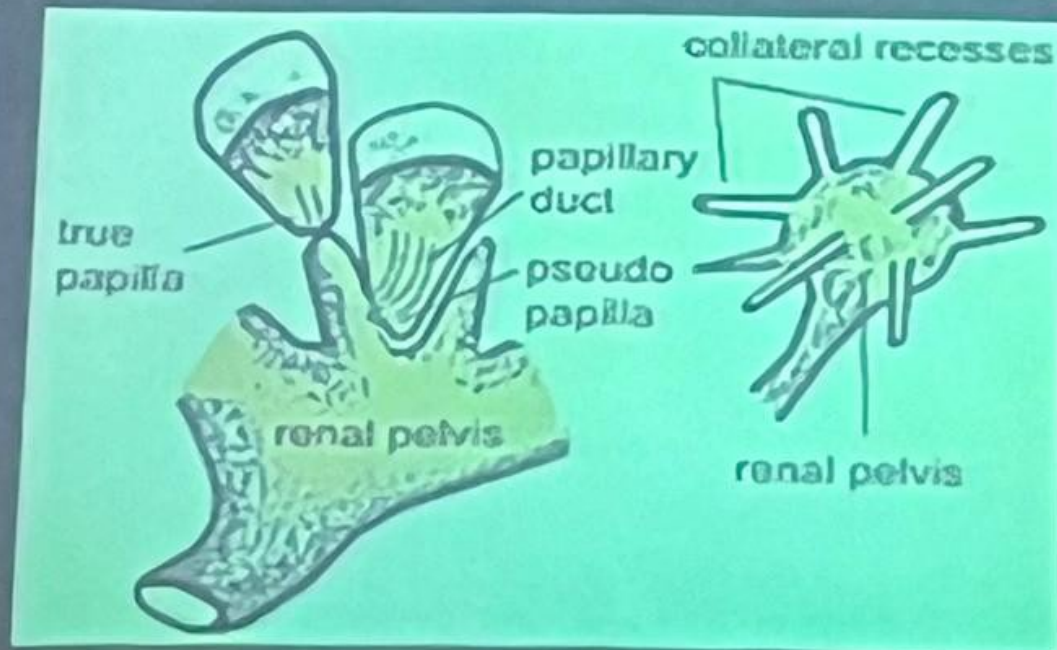
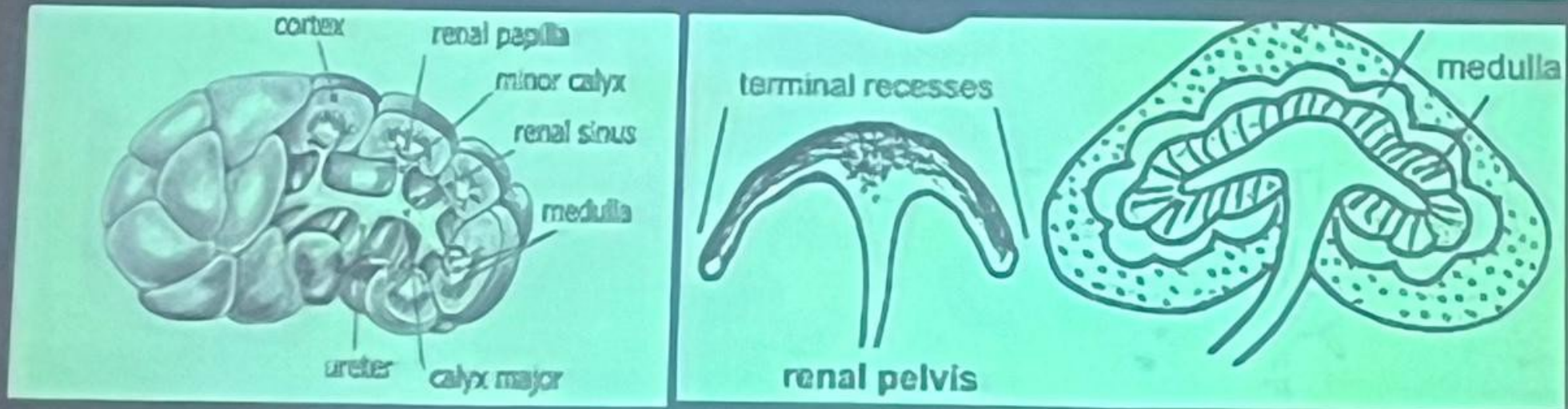
Renal hilus: A deep notch in the middle of the kidney's medial border through which the renal artery, renal vein, and ureter (renal pelvis) pass.



Renal pelvis is the dilated part of the ureter inside the kidney that collects urine from the renal papillae and transports it to the ureter.

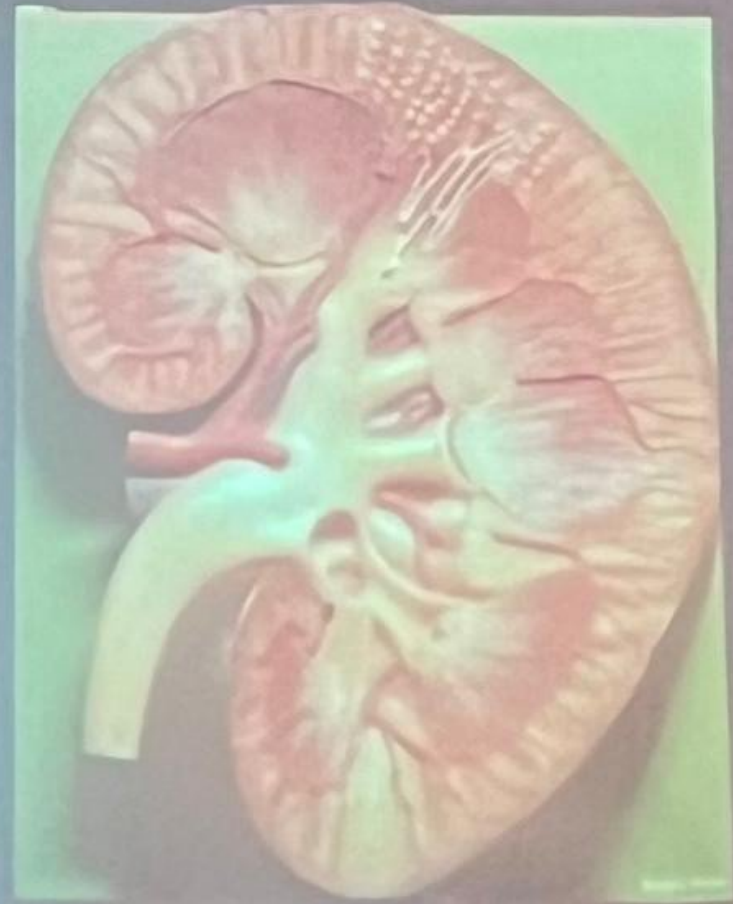
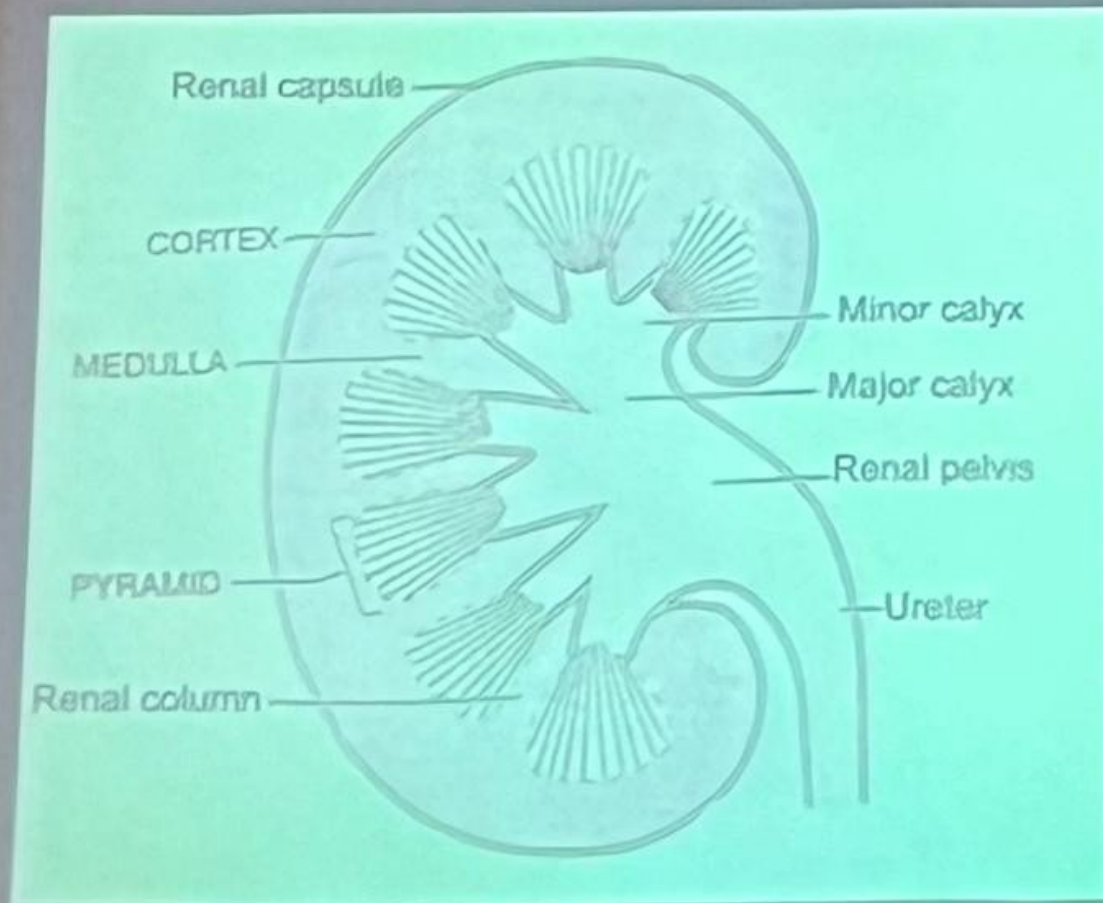


Ox	Horse	Dog & Camel
cavity divided into major & minor calyces	cavity & two lateral terminal recesses	cavity & many recesses (pseudo-multipapillary)



Cortex

- granular, reddish brown.
- Parts from cortex extend between renal pyramids called Columns of Bertin.



Fixation of kidney

1. **Pressure** of surrounding organs.
2. **Renal fascia & perirenal fat**; well developed in ruminants, moderate in dog, undeveloped in horse.
3. **Renosplenic ligament**: between left kidney and spleen.
4. **Hepatorenal ligament**: between right kidney and liver.

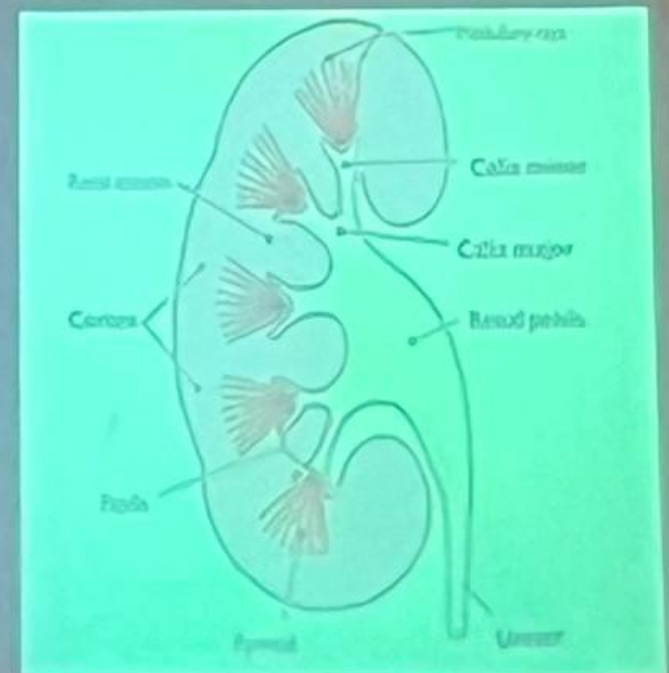
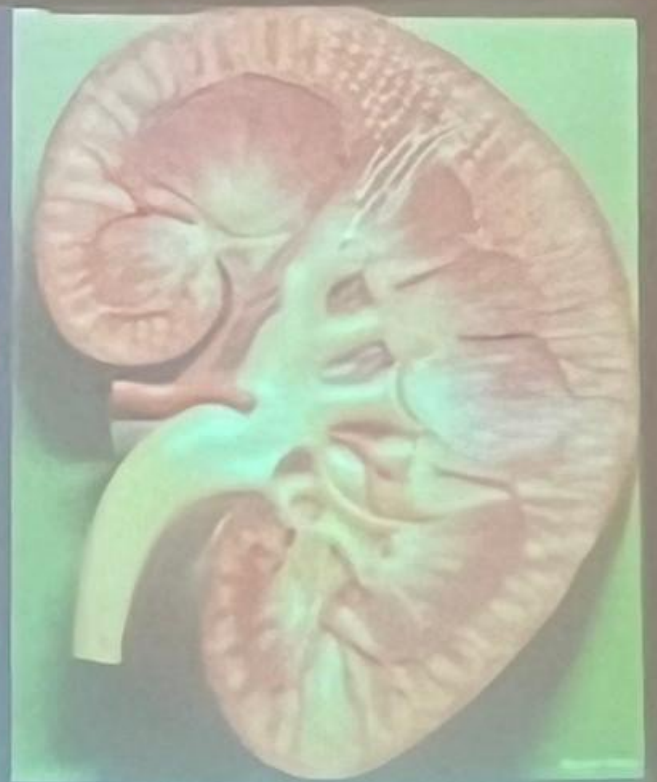
Medulla contains renal pyramids that contain a network of tubules.

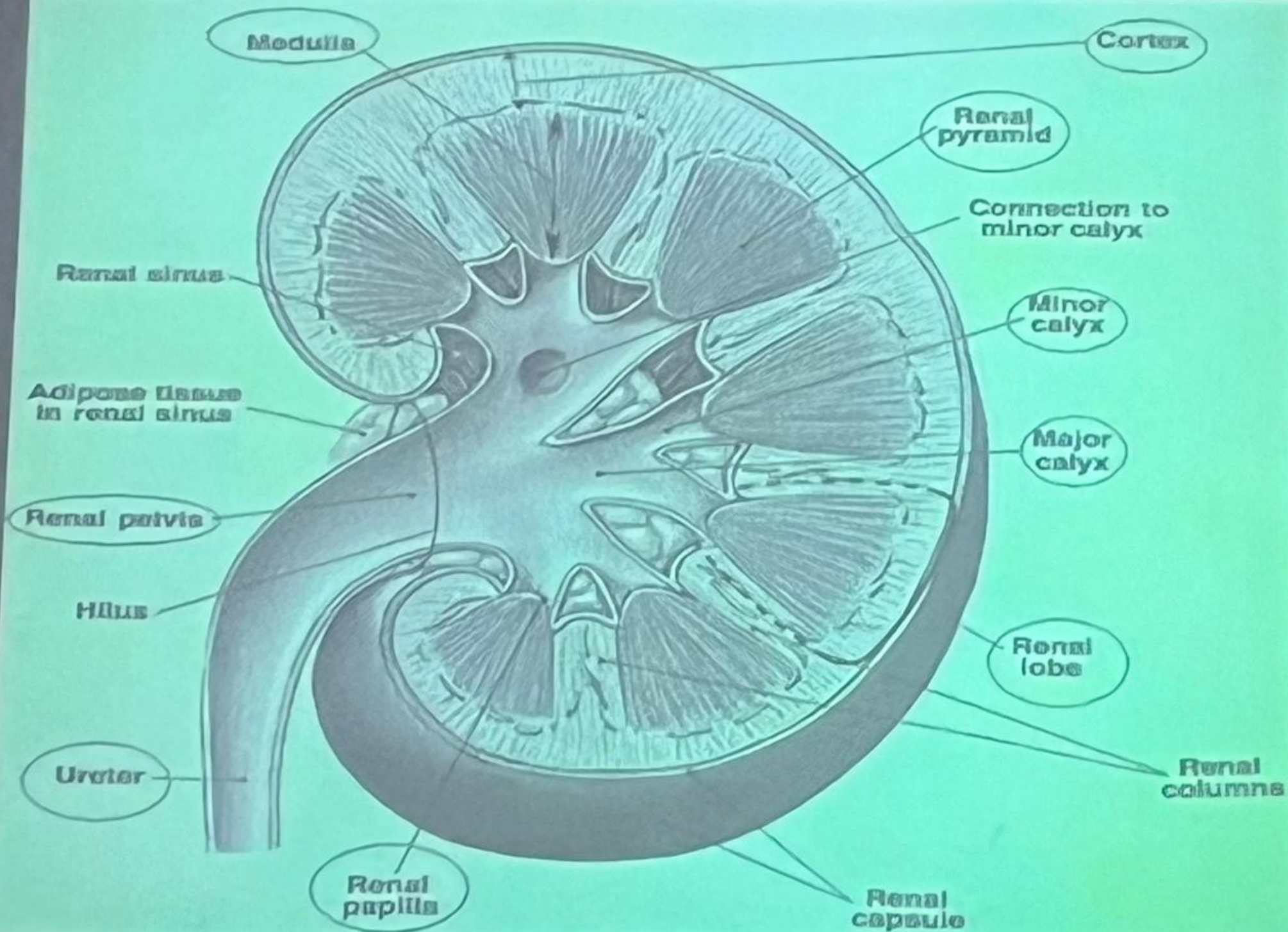
❖ **Medullary rays**: projections from base of medullary pyramids into cortex.

❖ **Renal papilla**: the tip of renal pyramid that drain urine to renal pelvis.

❖ **Renal lobe**: renal pyramid capped by part of cortex.

❖ **Renal lobule**: part of cortex which contains medullary rays and surrounded by interlobular arteries.



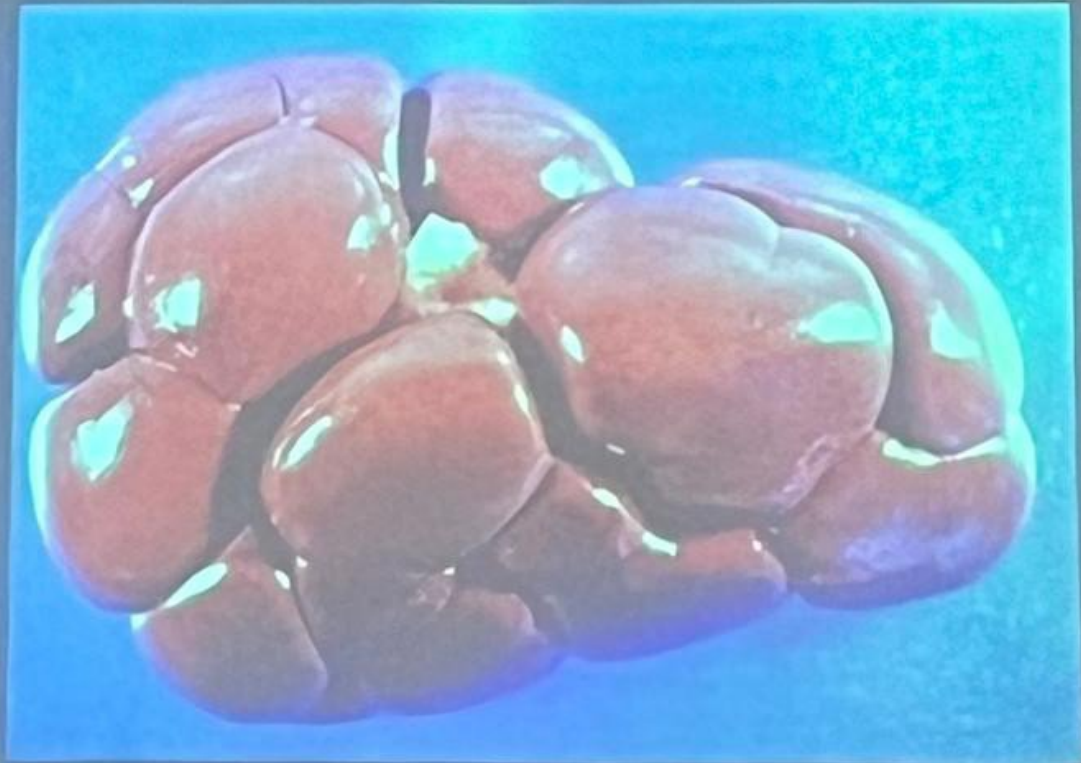


Classification (types) of kidney

❖ According to renal cortex

1- **Smooth kidney** (cortex completely fused) in all animals except in ox.

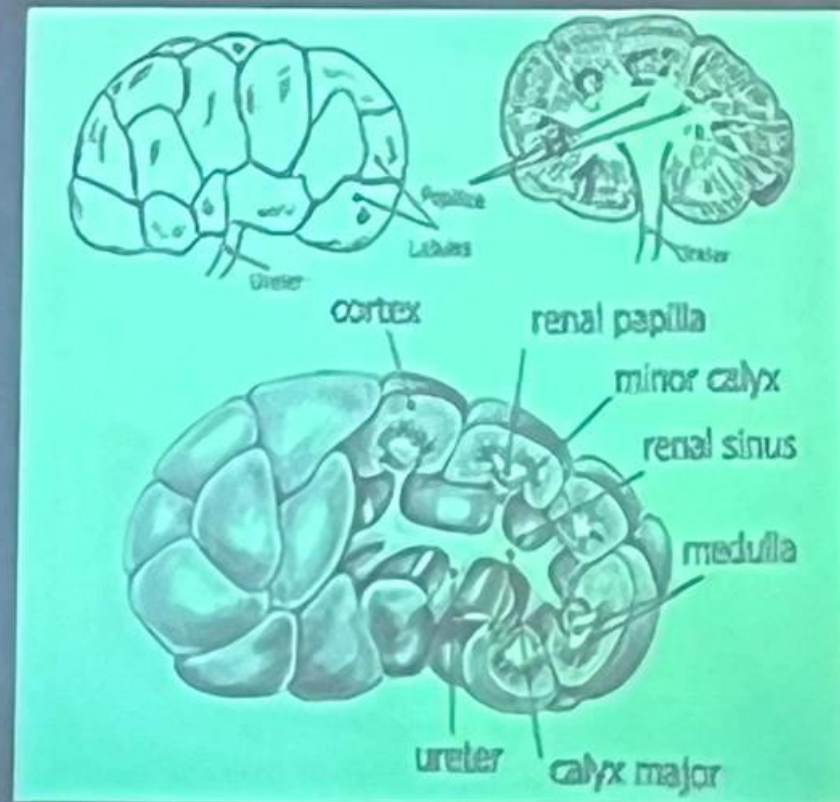
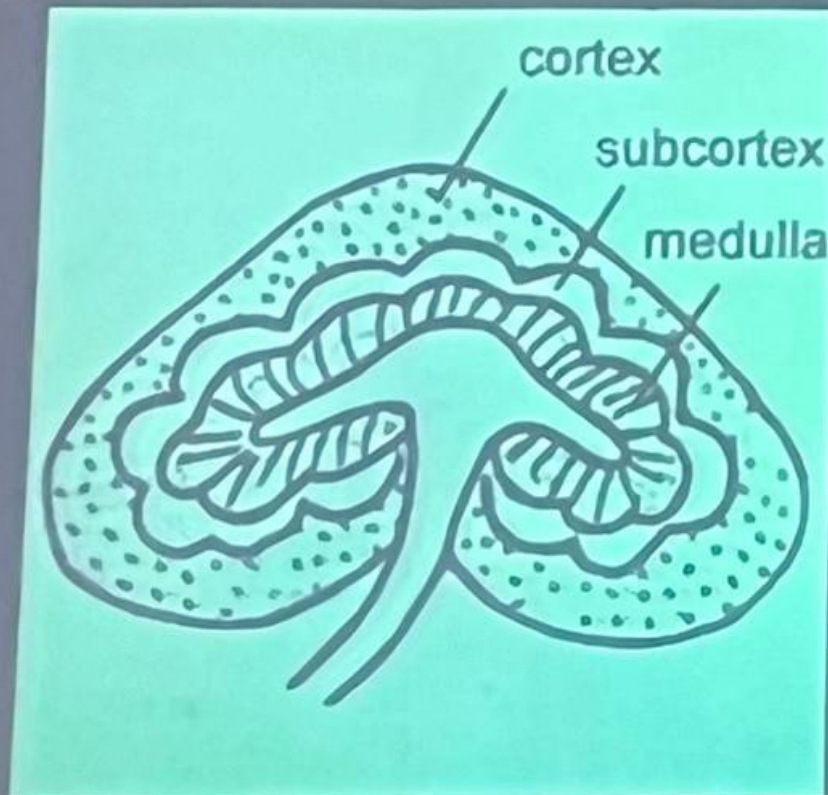
2- **Fissured (lobated) kidney** (cortex has several fissures) in ox.



❖ According to renal papillae

1- Unipapillary kidney (Renal papillae fused) in all animals except ox.

2- Multipapillary kidney (Many papillae each one opens into minor calix that collected into major calix) in ox.



- ❖ **wandering kidney:** left kidney of ox is free movable in abdominal cavity by long mesentery when the rumen is filled, it moved behind right kidney.
- ❖ **Left kidney in dog:** suspended by short mesentery so the position is affected by the degree of filling of stomach.
- ❖ The difference between the kidney of dog and sheep in lab is difficult.
- ❖ **Pseudo-multipapillary kidney** in dog and camel.

Blood supply of the kidney

- • **Abdominal aorta** gives renal artery.

Renal artery



Interlobar arteries between renal pyramids

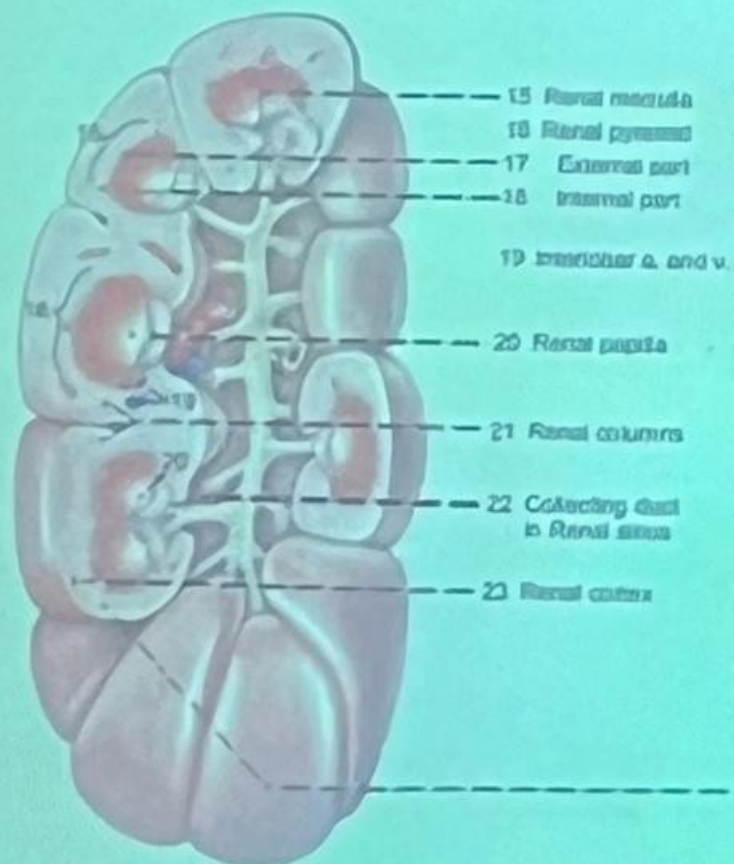


Arcuate arteries (junction between cortex and medulla)

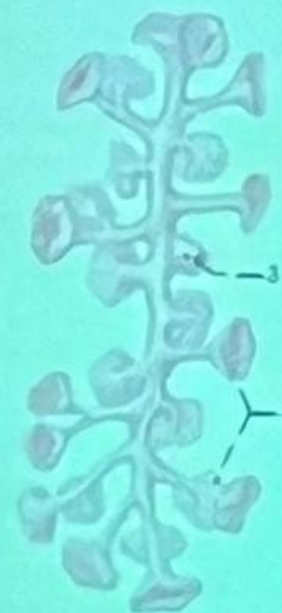


Interlobular arteries enter the cortex to supply the renal tubules and give off **afferent arteriole** enter glomeruli and **efferent arteriole** leave glomeruli).

Right kidney (Sectioned)



Ureter and Calices (Right kidney)



Left kidney



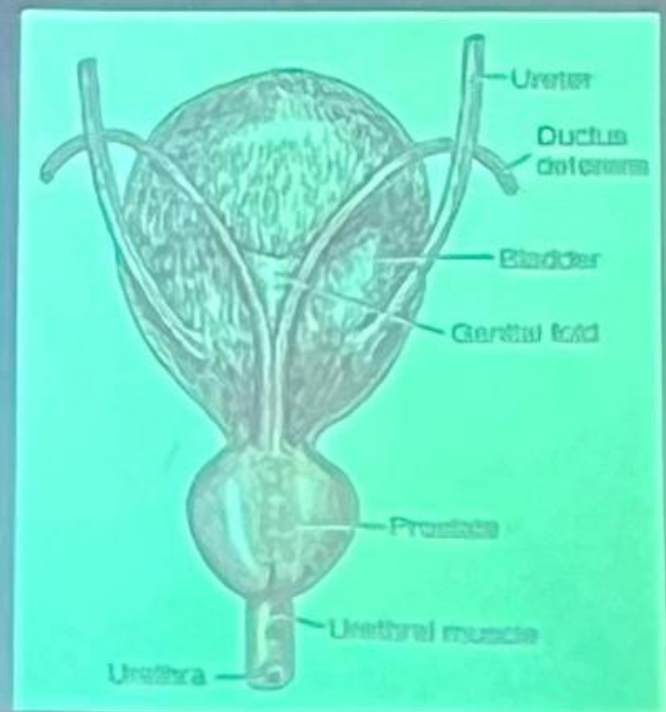
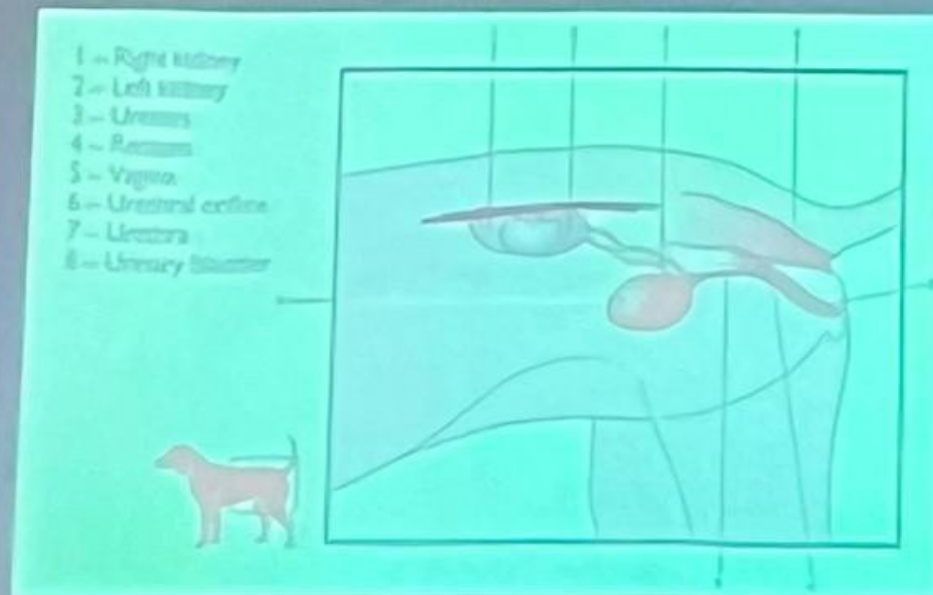
Kidney of ox

Vesical triangle (Trigonum vesicae)

- It is a smooth triangular area formed by the internal urethral opening and the two ureters' openings at the internal surface of UB situated near the neck of UB.
- It appears at empty UB where UB internal mucous membrane forms longitudinal folds except at trigonum vesicae.

Ureter

- a narrow musculo-membranous tube transmits urine from renal pelvis to urinary bladder.
- it courses retroperitoneal to enters the dorsal wall of the bladder near its neck.
- The proximal third of the ureter contain mucous glands (horse).



Kidney of camel

Shape

- Right kidney is bean shaped
- Left kidney is human ear shaped

Type

Smooth, Unipapillary, Multilobar

Position

- Asymmetric in position
- R. kidney: 2nd-5th lumbar vertebrae
- L. kidney: 5th-7th lumbar vertebrae

Anatomical description

- Capsula adipose: poor developed
- Fibrous capsule is thick
- Cortex dark brown, renal crest is long.
- Renal pelvis consist of:
 - small cavity
 - no/of collateral recess called renal column which run between renal pyramids so called pseudopapillary kidney

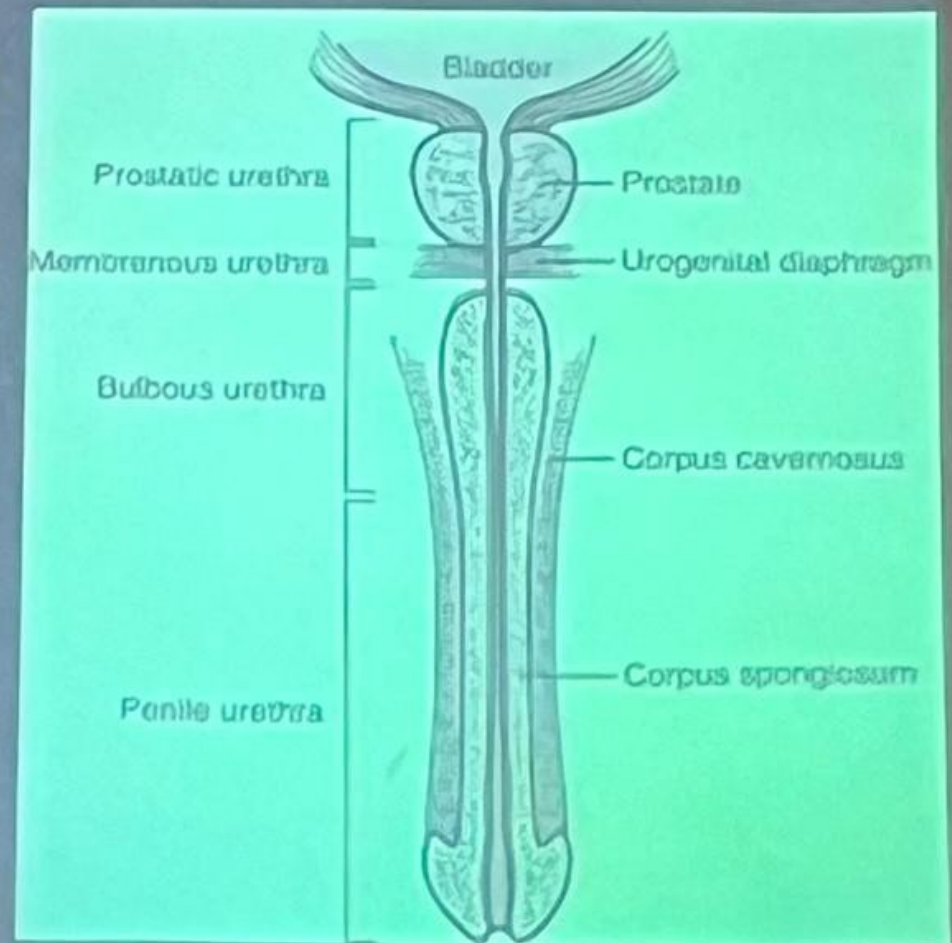
Male urethra

It is a musculomembranous tube extends from the internal urethral orifice at the neck of UB to the external urethral orifice at the free end of penis.

Parts of male urethra:

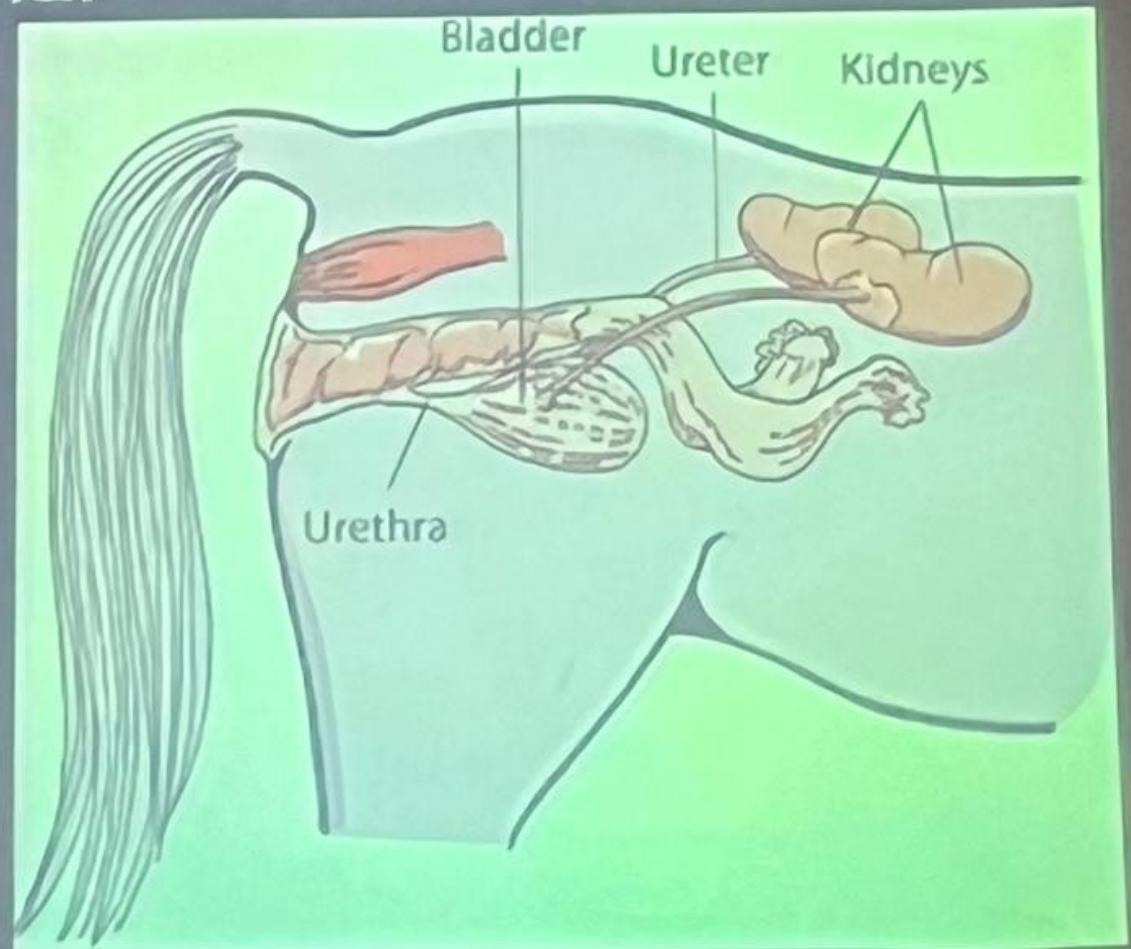
a. According to the product passing in urethra:

1. **Prostatic (prostatic) urethra** from internal urethral orifice to colliculus seminalis (urine).
2. **Urogenital canal** from colliculus seminalis to external urethral orifice (urine and semen).

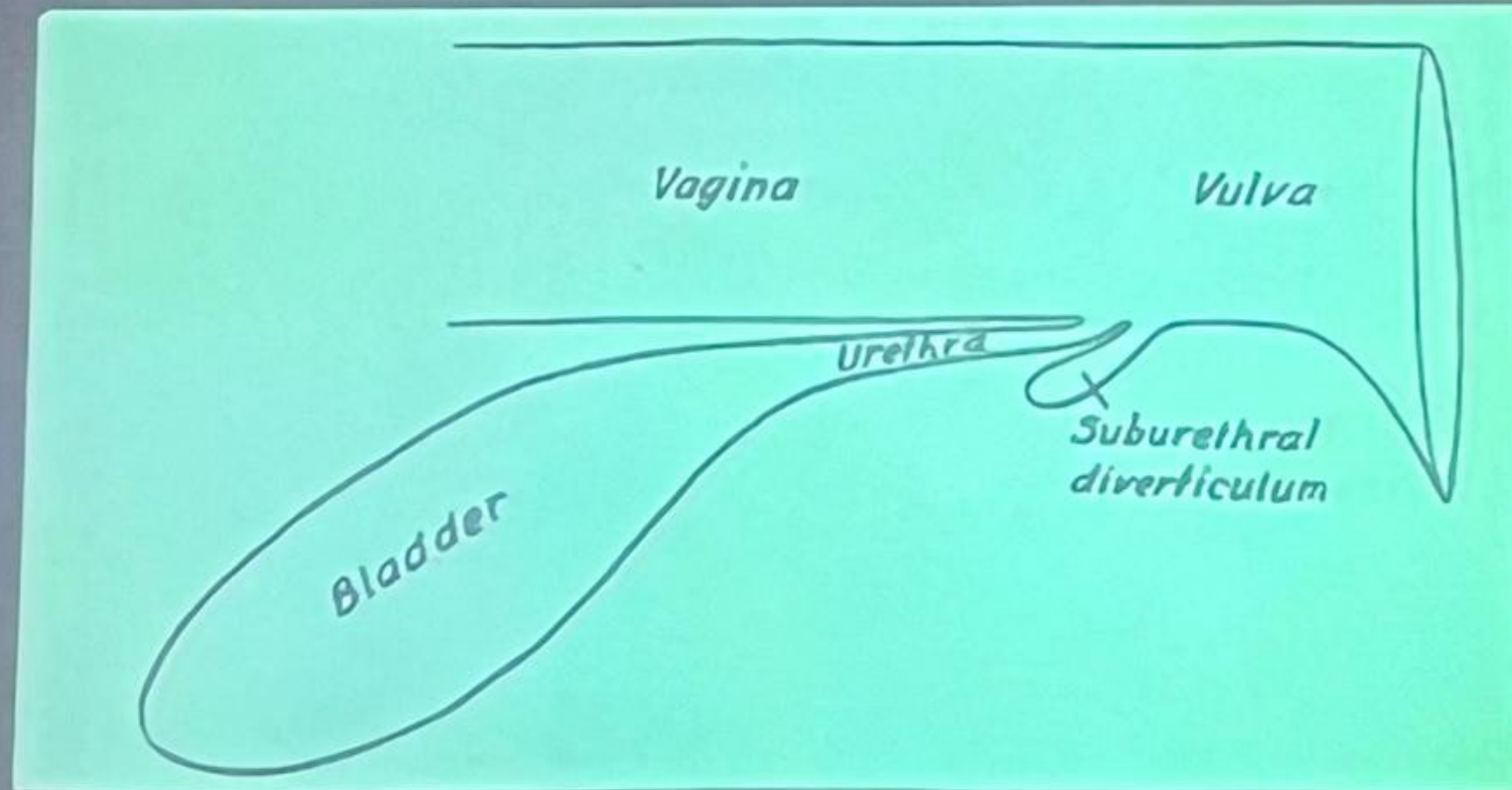


Female urethra

- it is called the urinary canal.
- It begins at the internal urethral orifice of the UB and ends at the external urethral orifice at the floor of the vagina.



A small blind diverticulum called **suburethral diverticulum** below the external urethral orifice. In cow and she-camel

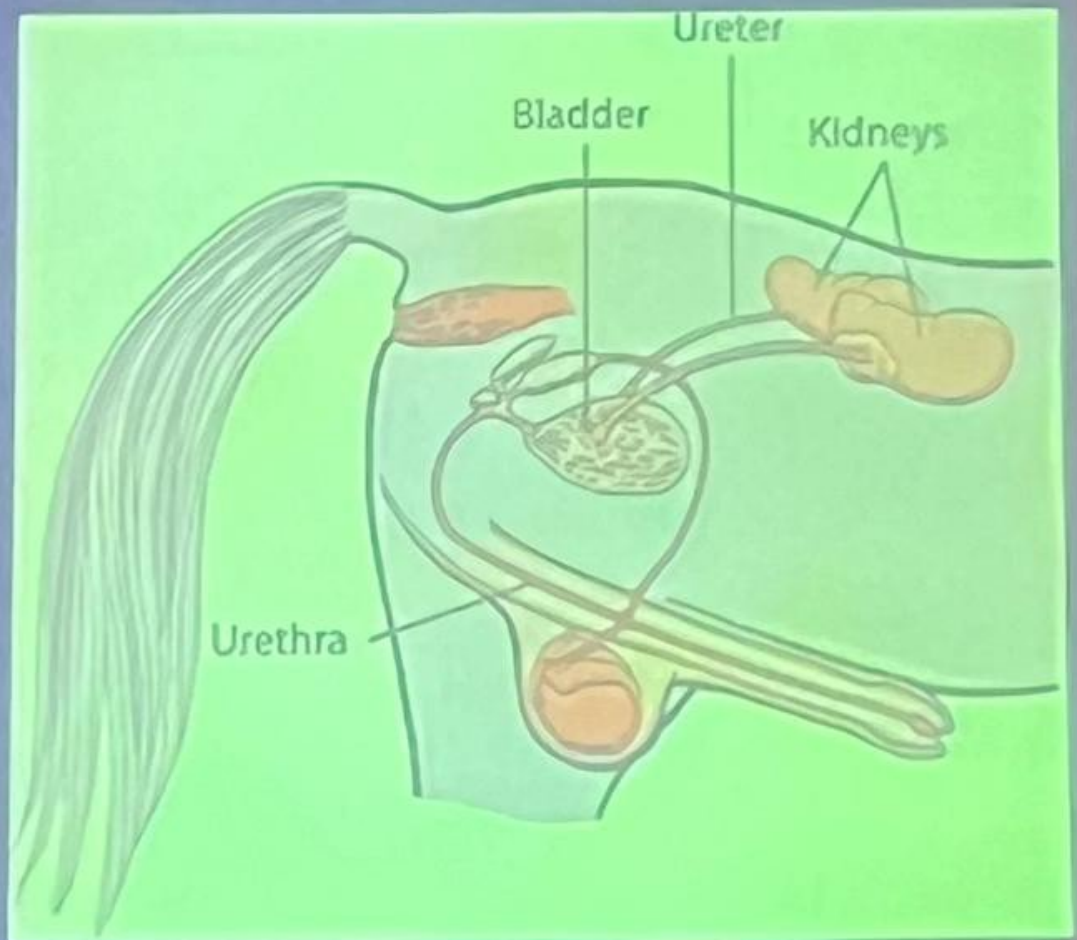


- In **bitch**: the external urethral orifice open into vagina on a small elevation flanked by two grooves.

b. According to the position of urethra:

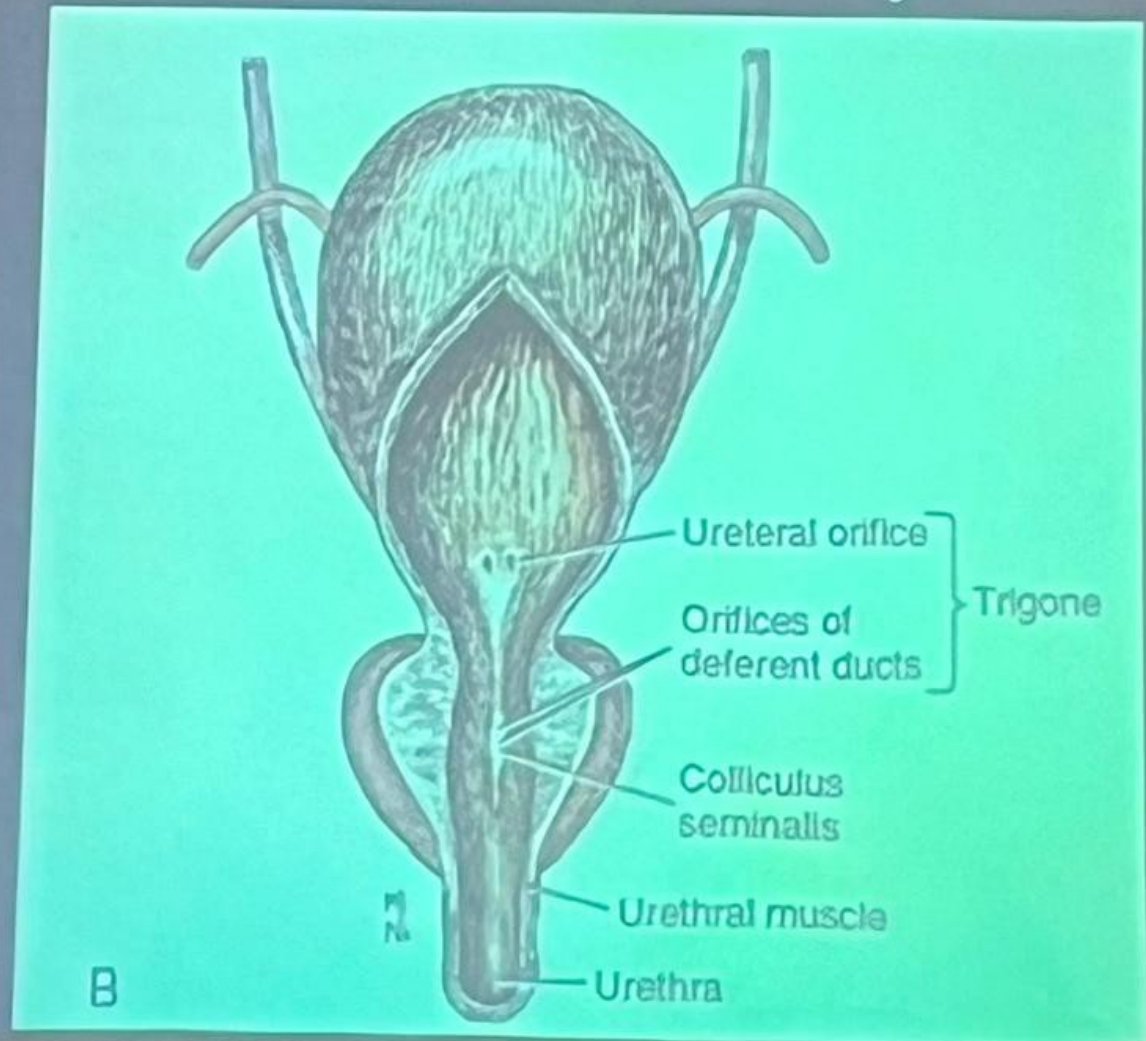
1. Pelvic part: from internal urethral orifice until ischial arch within the pelvic cavity.

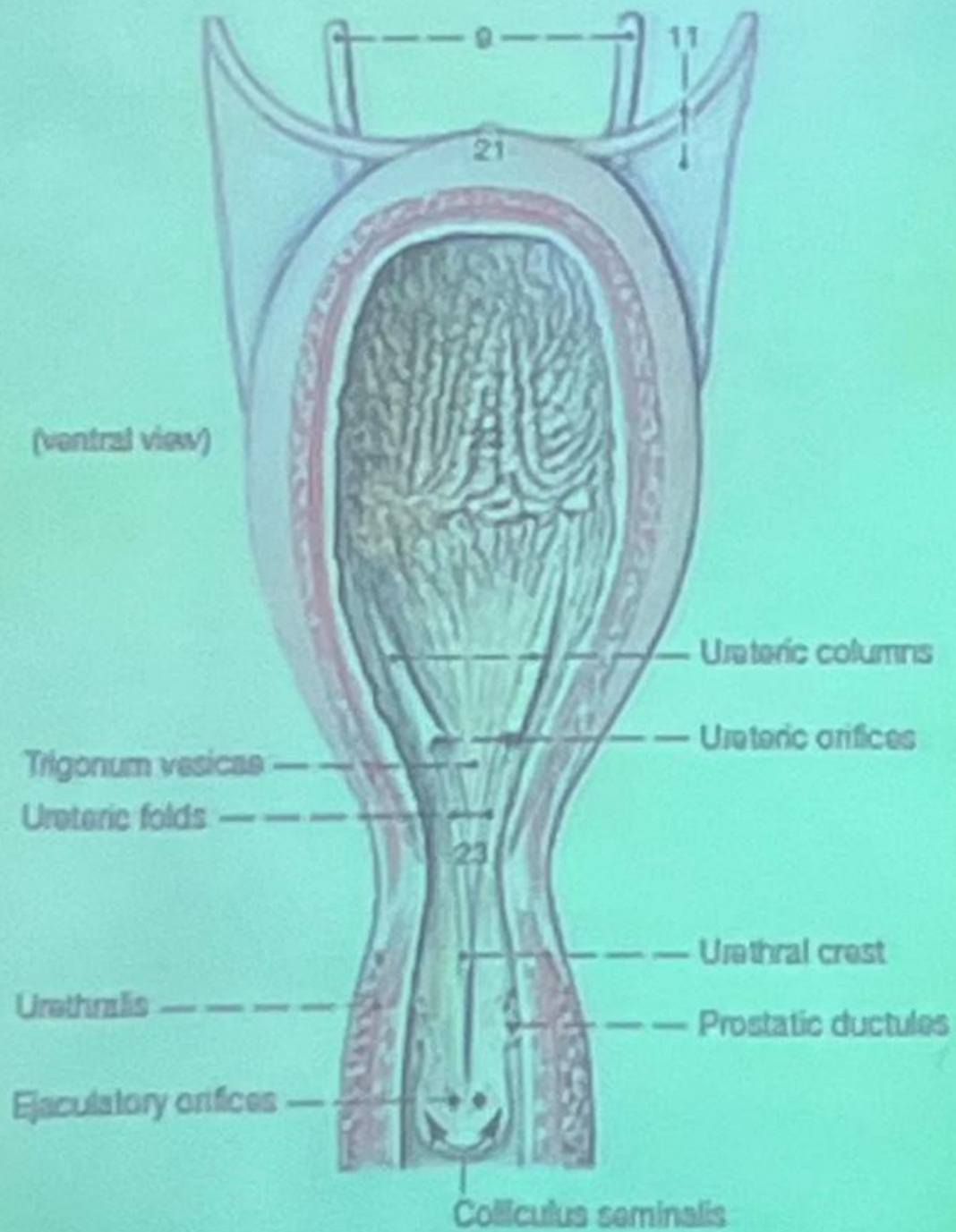
2. Penile part: from ischial arch to external urethral orifice within the penis.



Urinary bladder

- It consists of apex, neck, and body.





Fixation of the urinary bladder

It is fixed by three peritoneal folds:

- **Middle vesicular ligament** fixed UB to floor of pelvic cavity.
- **Two lateral vesicular ligaments** fixed UB to lateral walls of pelvic cavity.
- **Round ligament** is a firm band at the free end of lateral vesicular ligament (remnant of umbilical artery).

Kidney of equine

Shape

- Right kidney is heart
- Left kidney is bean

Type

Smooth, Unipapillary, Multilobar

Position

- R. kidney: 16th rib-1st lumbar vertebrae.
- L. kidney: 17th rib –2nd lumbar vertebrae

Anatomical description

- Capsula adipose: poor developed
- Renal hilus is deep, common renal papilla crest like projected into the central cavity of renal pelvis
- Renal pelvis consists of small central cavity and two lateral terminal recesses.

Kidney of small ruminants

Shape

Round bean

Classification

Smooth, Unipapillary, Multilobar

Position

- R. kidney: 12th rib–2nd lumbar vertebrae
- L. kidney: 2nd –4th lumbar vertebrae. it was mobile, and its position varied according to the degree of fullness of the rumen due to its long mesentery.

Anatomical description

- Capsula adipose: well developed
- The difference between the kidney of dog and sheep is difficult.
- Renal pelvis consists of:
 - ✓ small cavity and connected to 10-16 collateral recess called renal column which run between renal pyramids so called pseudopapillary kidney

Kidney of ruminants

Shape

- Right kidney is oval.
- Left kidney is pyramidal.

Type

Fissured, Multipapillary, Multilobar

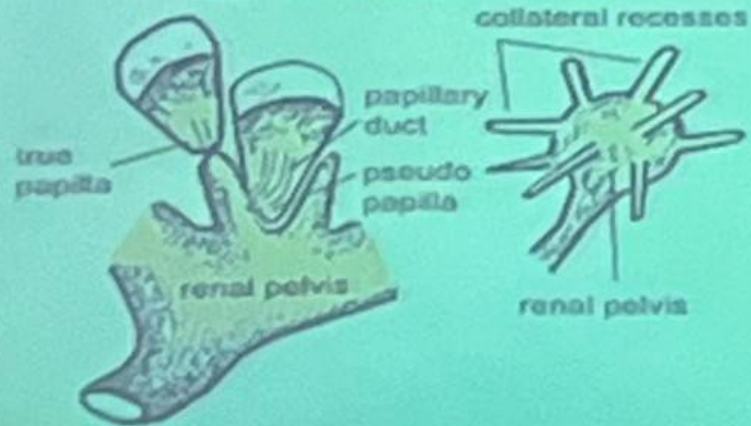
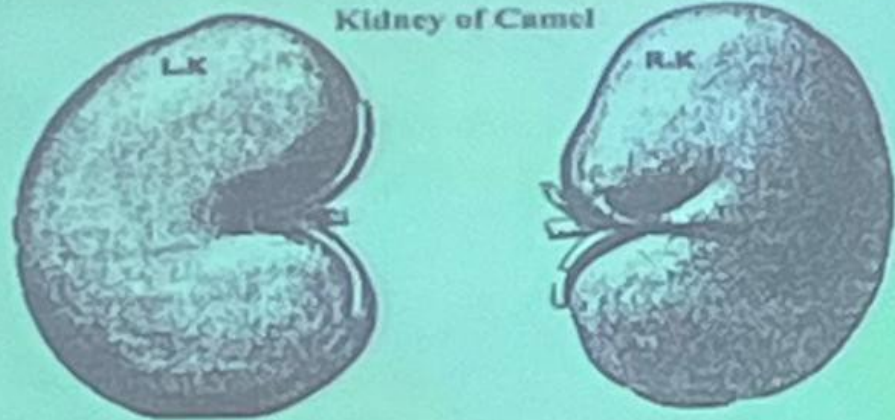
Position

- R. kidney: last rib–3rd lumbar vertebrae
- L. kidney: 2nd –5th lumbar vertebrae: wandering kidney, it is hanged freely in abdominal cavity by peritoneal fold. When rumen is filled, left kidney move in pendulous movement to right side.

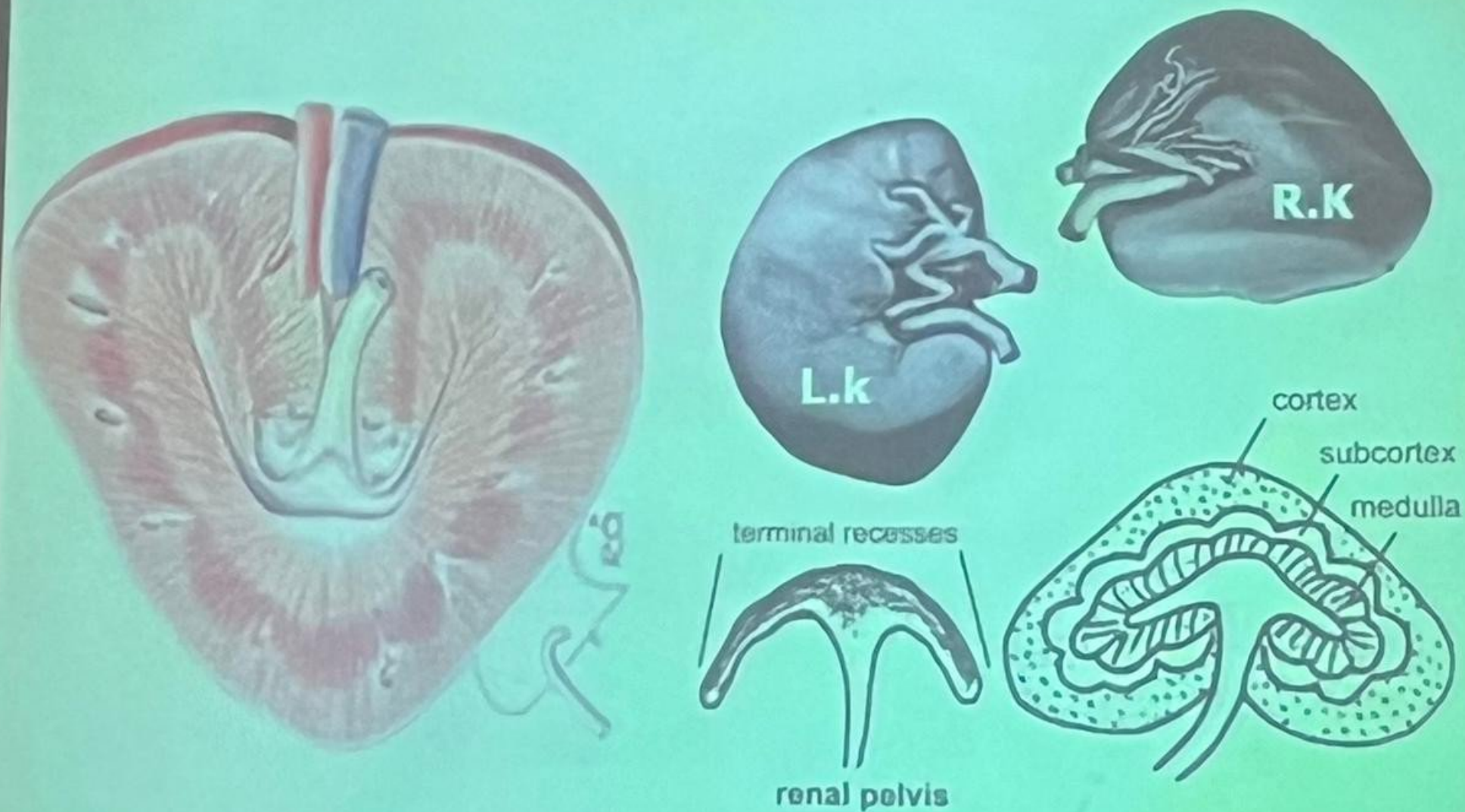
Anatomical description

- Capsula adipose: well developed
- Renal pelvis: is absent
- A minor calyx surrounds the renal papillae of each pyramid and collects urine from that pyramid. Several minor calyces converge to form a major calyx. From the major calyces, the urine flows into the renal pelvis; and from there, it flows into the ureter.
- Renal pyramids and column are clear

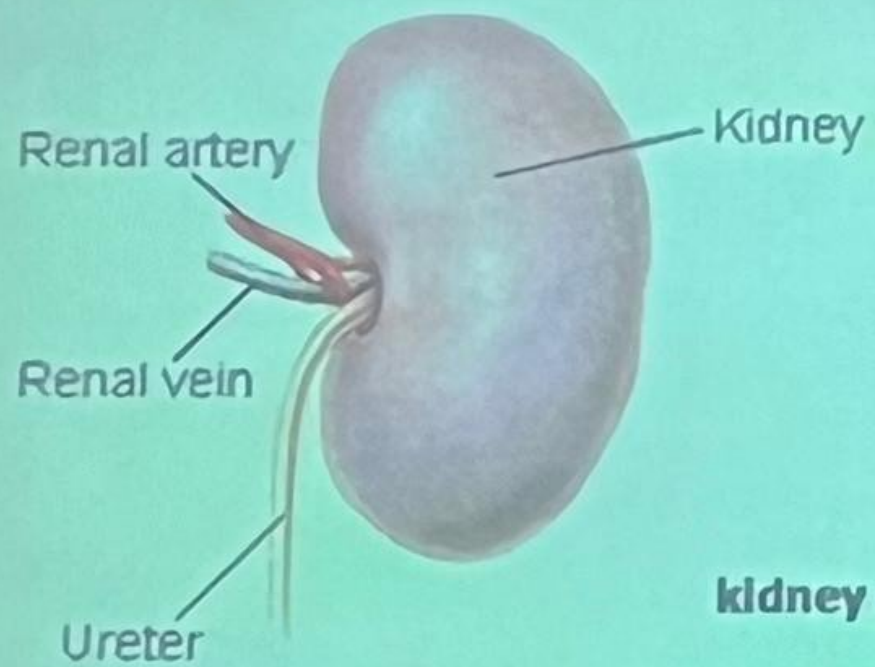
Kidney of Camel



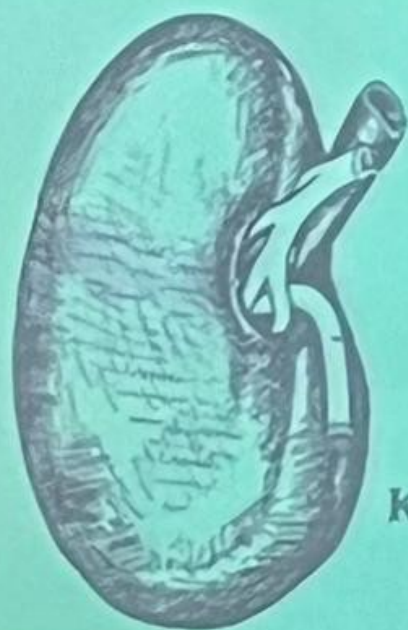
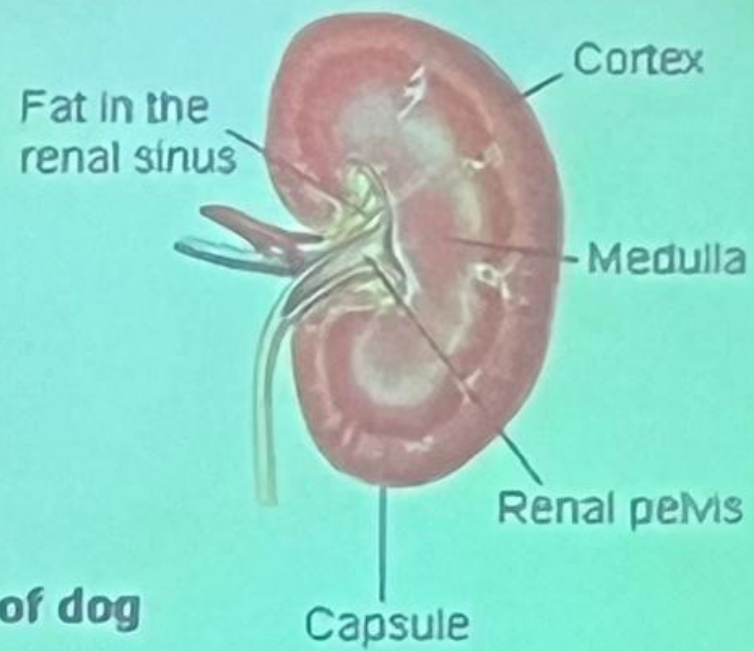
	Kidney of dog
Shape	Irregular bean
Type	Smooth, Unipapillary, Multilobar
Position	• Asymmetric in position • R. kidney: last intercostal space to 3rd lumbar vertebrae. • L. kidney: 2nd - 4th LV suspended by peritoneum affected by filling of stomach
Anatomical description	• Capsula adipose: less developed • Renal pelvis consist of: ✓ small cavity ✓ no/of collateral recess called renal columns which run between renal pyramids so called pseudopapillary kidney.
Other	Can be palpated through abdominal wall



Kidney of horse



kidney of dog



Kidney of Sheep

